

PHYS 206 Experiment 5 Lab Report: Bohr Theory of Hydrogen - The Rydberg Constant

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1 Introduction

In this experiment, the atomic emission spectra of different atomic discharge lamps are investigated. The primary objective is to calibrate the student spectrometer using the well-known atomic transition lines of mercury (Hg) and subsequently utilize this calibration to measure the atomic transitions of hydrogen (H). By analyzing the characteristic wavelengths of the hydrogen line spectrum, the experimental value of the Rydberg constant can be determined and compared with the theoretical framework established by the Bohr model of the atom.

2 Theory

When electrons in an atom drop from a higher energy state to a lower energy state, they emit electromagnetic radiation in the form of photons. This process produces an emission spectrum characteristic of the specific element. In a student spectrometer setup, a diffraction grating can be utilized to separate this emitted light into its component wavelengths by causing constructive interference at specific angles.

The relationship between the wavelength (λ) of the light, the grating spacing

(d), and the diffraction angle (θ) is governed by the grating equation:

$$\lambda = d \sin(\theta) \quad (1)$$

By measuring the diffraction angles for known spectral lines of a Mercury lamp, the grating spacing d can be precisely calibrated.

According to the Bohr model and the Rydberg formula for hydrogen, the inverse of the wavelength (λ^{-1}) for transitions terminating at the $n_f = 2$ energy level (the Balmer series) is related to the principal quantum number of the initial state (n_i) by:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{n^2} \right) = R_H \left(\frac{1}{4} - \frac{1}{n^2} \right) \quad (2)$$

where R_H is the Rydberg constant and n represents the upper energy level ($n = 3, 4, 5, 6$). By plotting λ^{-1} against $\left(\frac{1}{4} - \frac{1}{n^2}\right)$, a linear fit can be performed where the slope directly yields the experimental value of R_H .

3 Setup

3.1 Equipment

- Student spectrometer (used with a diffraction grating with 600 lines/mm)
- Atomic discharge lamps (Mercury (Hg) and Hydrogen (H))
- High voltage power supplies



Figure 1: Placeholder for Spectrometer Setup Diagram

4 Procedure

1. Set up the student spectrometer with the diffraction grating and align it with the high-voltage Mercury (Hg) atomic discharge lamp.
2. Measure the diffraction angle (θ) for the prominent violet, blue, and green emission lines of the Hg lamp twice (on both sides or independent trials) to calibrate the grating spacing (d).
3. Replace the Mercury lamp with the Hydrogen (H) discharge lamp without changing the alignment configuration.
4. Measure the diffraction angles (θ) twice for the visible Balmer series lines corresponding to different quantum numbers ($n = 3, 4, 5, 6$).
5. Use the calibrated grating equation to determine the experimental wavelengths and perform a linear fit to extract the Rydberg constant.

5 Data Analysis

In our spectrometer, we measure the center of the pattern at $\theta = 185^\circ$, so I used $\theta_{measured} - 185^\circ$ as θ_1 and θ_2 .

5.1 Grating Calibration using Mercury (Hg)

Using the grating equation $\lambda = d \sin(\theta)$, the grating spacing d is calculated from the measured angles of known mercury lines.

Table 1: Mercury Calibration Data and Grating Spacing Determination

Color	Intensity	λ (Å)	θ_1	θ_2	θ_{avg}	Calculated d (Å)
Violet	400	4046.563	14° 12'	15° 12'	14° 42'	15946.55
Blue	1000	4358.328	15° 18'	15° 18'	15° 18'	16516.76
Green	500	5460.735	19° 18'	19° 24'	19° 21'	16480.86

Average Grating Spacing (d_{avg}): 16314.72 Å

5.2 Hydrogen (H) Spectrum Measurements

The table below displays the measured diffraction angles for the hydrogen discharge lamp lines and using the average grating spacing d_{avg} , the experimental wavelengths are computed and compared against the literature values.

Table 2: Hydrogen Line Spectrum Angular Measurements

n	Color	Intensity	λ (Å)	θ_1	θ_2	θ_{avg}	λ_{calc} (Å)	% Error
6	Violet	15	4101.7400	14° 36'	14° 36'	14° 36'	4112.44	0.26%
5	Blue	30	4340.4620	16° 12'	16° 12'	16° 12'	4551.66	4.87%
4	Turquoise	30	4861.2786	17° 54'	17° 54'	17° 54'	5014.44	3.15%
4	Turquoise	10	4861.2870	17° 00'	17° 00'	17° 00'	4769.96	1.88%
4	Turquoise	60	4861.3615	17° 06'	17° 06'	17° 06'	4797.19	1.32%
3	Red	90	6562.7110	21° 30'	21° 30'	21° 30'	5979.37	8.89%
3	Red	30	6562.7248	22° 24'	22° 24'	22° 24'	6217.06	5.27%
3	Red	180	6562.8518	23° 12'	23° 12'	23° 12'	6427.05	2.07%

5.3 Determination of the Rydberg Constant

To prepare for the linear fit (λ^{-1} vs. $1/4 - 1/n^2$), the corresponding inverse values are calculated using the original literature wavelengths ($\lambda_{Literature}$) provided in the lab manual.

Table 3: Inverse Wavelength and Quantum Number Terms for Linear Fitting

n	Color	λ (Å)	λ^{-1} (10^{-4} Å $^{-1}$)	$(\frac{1}{4} - \frac{1}{n^2})$
6	Violet	4101.7400	2.4380	0.2222
5	Blue	4340.4620	2.3039	0.2100
4	Turquoise	4861.2786	2.0571	0.1875
4	Turquoise	4861.2870	2.0571	0.1875
4	Turquoise	4861.3615	2.0570	0.1875
3	Red	6562.7110	1.5238	0.1389
3	Red	6562.7248	1.5238	0.1389
3	Red	6562.8518	1.5237	0.1389

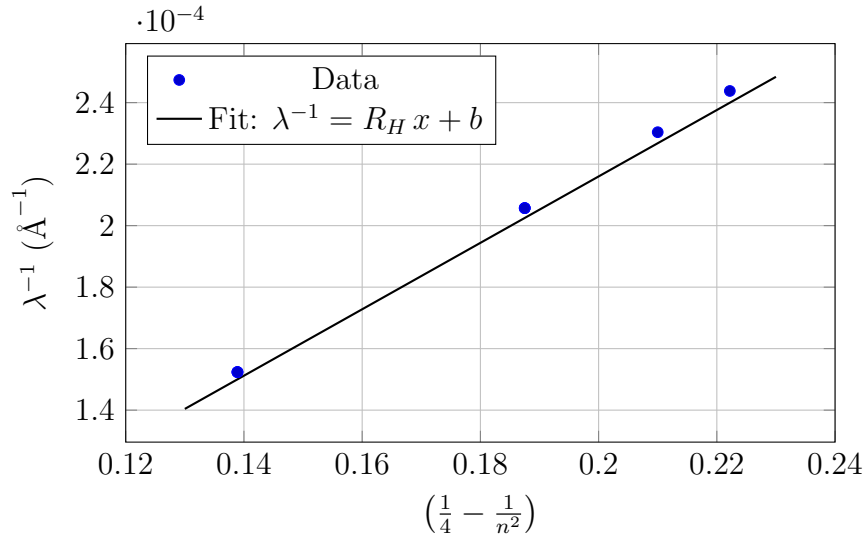


Figure 2: Linear fit of λ^{-1} versus $(\frac{1}{4} - \frac{1}{n^2})$.

From the slope of the linear fit line, the experimental Rydberg constant is found to be:

$$R_H = 0.00108 \text{ Å}^{-1}$$

6 Questions

1. **Where do we use atomic mercury (Hg) lamps?**

Answer: Atomic mercury lamps are used in commercial fluorescent tubes for laboratories and classrooms, where its emission excites a phosphor coating to produce visible white light. They are used for UV sterilization.

2. **Where do we use atomic hydrogen (H) lamps?**

Answer: Atomic hydrogen lamps are used as stable, continuous light sources for UV spectroscopy in chemistry. They are also used in laboratory setups to study fundamental quantum mechanics and analyze atomic emission structures.

3. **What is the source of the %error in the wavelength measurements?**

Answer: The error could be due to human reading precision, as the spectroscope is difficult to read accurately, and aligning the colored line is difficult. Additionally, misalignment between the lamp and the slit can also cause to the error.

4. **The quantum number (n) belongs to which physical variable?**

Answer: The principal quantum number (n) corresponds to the energy level of the electron within the atom.

5. **What is the name of the line series observed in hydrogen (H) measurements?**

Answer: The line series observed in these measurements is the Balmer series.

7 Discussion

In this experiment, we calibrated the spectrometer using a mercury discharge lamp to find an average grating spacing, which was subsequently used to calculate the wavelengths of the hydrogen Balmer series. The experimental results showed good agreement with theoretical values, with the percentage error ranging from a minimal 0.26% for the violet line ($n = 6$) to a maximum of 8.89% for the red line ($n = 3$). These errors are based on human limitations in aligning the center of the spectral lines, precision of reading the the

angle, and optical misalignments between the lamps and the slit. Despite these small experimental errors, the linear relationship observed in our data confirms the fundamental energy level quantization predicted by the Bohr model.